



Arvin Aryanpour

CONTACT

647-919-1103
arvinaryanpoura@gmail.com
linkedin.com/in/arvinary
github.com/TheArvinA

SKILLS & CERTIFICATES

- **Programming:** Python, C/C++, Java, JavaScript, C#, SQL, VHDL, MATLAB
- **AI/ML & Computer Vision:** PyTorch, YOLOv8, OpenCV, ByteTrack, Scikit-learn, NumPy, Reinforcement Learning (PPO, Unity ML-Agents)
- **Backend & Systems:** FastAPI, REST APIs, Microservices, PostgreSQL, SQLite, WebSockets
- **Frontend:** React, HTML/CSS, UI/UX Integration
- **Tools & Platforms:** Git, Linux/UNIX, FPGA (Quartus), Arduino, Unity
- **Core Strengths:** System Design, Embedded Systems, Computer Vision, OOP, Agile/Sc

LANGUAGE

Python, Java, JavaScript, C, C++, C#, SQL, HTML/CSS, MATLAB, VHDL

EDUCATION

Toronto Metropolitan University
Bachelor of Engineering in Computer Software Engineering

SOFTWARE ENGINEER

SUMMARY

Software Engineer specializing in AI-driven systems, computer vision, and full-stack development, with hands-on experience building scalable pipelines from data processing to real-time applications. Strong foundation in machine learning, backend architecture, and system design, with the ability to translate complex technical concepts into efficient, user-focused solutions. Experienced across hardware, backend, and frontend, with a focus on performance, scalability, and practical impact.

PROJECTS & ACHIEVEMENTS

- Built YOLOv8 + ByteTrack computer vision pipeline (87.9% precision) and full-stack video system (FastAPI + React) for real-time volleyball analytics
- Designed PPO reinforcement learning agent in Unity ML-Agents and FPGA processor (VHDL) with full simulation validation
- Engineered microservices-based smart parking system with dynamic pricing and scalable backend APIs
- Delivered end-to-end AI systems from training to deployment, demonstrating technical breadth across computer vision, ML, embedded systems, and full-stack development

EXPERIENCE

Sporting Life

Footwear Sales Associate / Sep. 2019 – Present

- Ranked #1 in department sales by applying data-driven decision-making and analyzing customer behavior patterns
- Translated complex product specifications into clear, user-focused recommendations that consistently exceeded KPIs
- Adapted strategies in real time based on feedback, demonstrating iterative problem-solving in high-volume environments

Pro Insulation Contracting

Web Developer– Dec 2024– Jan. 2026

- Built lead-generation web applications using FastAPI and REST APIs, improving conversion rates and user engagement
- Optimized performance and SEO, reducing load times and increasing inbound traffic
- Integrated analytics pipelines to track user behavior and marketing performance metrics
- Translated business requirements into scalable technical solutions across frontend and backend systems

VerticalScope

Embedded Software Engineer Intern – Sep. 2020– Apr. 2021

- Designed and implemented embedded systems using C/C++ firmware for real-time, constrained environments with high reliability requirements
- Integrated software with hardware systems, improving stability, execution efficiency, and overall system behavior
- Conducted testing, validation, and performance profiling to identify bottlenecks and implement optimizations
- Applied system-level design principles to ensure scalability, maintainability, and clean architecture